Finite terms (`Other_terms.tex`): why the cross section is real

1 What can be complex, and what is real

Real by construction:

- adjoint Wilson lines $U^{ab}(x)$ and structure constants f^{abc};
- $\rho^a(x)$ is hermitian (real if treated classically);
- all transverse kernels (rational functions of $(x-y)^i$, $(x-y)^2$);
- $\mathcal{A}_0$: its radicand $(x-w)^2(x-z)^2-[(x-w)\cdot(x-z)]^2\geq 0$ (Cauchy–Schwarz), and arctan of a real number is real;
- every logarithm whose argument is positive. In particular $L_x\equiv\log\frac{(1-\kappa)(x-z)^2+\kappa(x-w)^2}{(x-w)^2}$, with $\kappa\equiv k^+/\vee\in(0,1)$, has a positive argument (a convex combination of squares).

Possible sources of i:

(C1) the explicit i in the three-ρ terms (ig^4f^{abd}, $ig^4f^{c'd'a}$, ...);

(C2) the Fourier phases $e^{-ik\cdot(w'-w)}$, $e^{-ik\cdot(y'-w)}$, $e^{-ik\cdot(z-x)p^+/k^+}$;

(C3) the momentum integral in N_{2b} of Group I, Row II (an implicit i, Sec. 3);

(C4) logarithms with sign-changing arguments. These occur *only* in Group I, Row III (Sec. 6);

(C5) the ordering of ρ's, only if ρ is an operator (Sec. 2.3).

2 The rule: hermitian conjugation

$d^3N/d^3k=\langle\Psi|\hat{N}_k|\Psi\rangle$ with $\hat{N}_k$ hermitian.

- Groups I and III are interferences $\mathcal{A}_{\rm LO}^\dagger\mathcal{A}_{\rm NLO}$. Each appears together with its hermitian conjugate ("c.c.").
- Group II is $|\mathcal{A}|^2$ of the two-gluon final state. It has no separate c.c.; it must be hermitian by itself.

The conjugation † complex-conjugates every c-number and reverses the order of ρ's. A generic term reads

$$T=\int e^{-ik\cdot(w'-w)}\,c\,\mathcal{K}\,\mathcal{O},\qquad T^\dagger=\int e^{+ik\cdot(w'-w)}\,c^*\,\mathcal{K}\,\mathcal{O}^\dagger,\tag{2.1}$$

with c a constant, $\mathcal{K}$ a real kernel and $\mathcal{O}$ the colour/ρ structure.

2.1 Class R (real coefficient c)

$$T+T^\dagger=\int\mathcal{K}\big[e^{-ik\cdot(w'-w)}\mathcal{O}+e^{+ik\cdot(w'-w)}\mathcal{O}^\dagger\big]\ \overset{\rho\ \text{classical}}{=}\ 2c\int\cos\big(k\cdot(w'-w)\big)\,\mathcal{K}\,\mathcal{O}.\tag{2.2}$$

2.2 Class I (imaginary coefficient $c=i\alpha$, α real)

$$T+T^\dagger=\alpha\int\mathcal{K}\,i\big[e^{-ik\cdot(w'-w)}\mathcal{O}-e^{+ik\cdot(w'-w)}\mathcal{O}^\dagger\big]\ \overset{\rho\ \text{classical}}{=}\ 2\alpha\int\sin\big(k\cdot(w'-w)\big)\,\mathcal{K}\,\mathcal{O},\tag{2.3}$$

because $i\,e^{-ik\cdot\Delta}+(i\,e^{-ik\cdot\Delta})^*=2\sin(k\cdot\Delta)$. **This is how every explicit i disappears:** the pair "$i\,e^{-ik\cdot\Delta}$ + c.c." becomes the real $2\sin(k\cdot\Delta)$, and "$e^{-ik\cdot\Delta}$ + c.c." becomes $2\cos(k\cdot\Delta)$.

2.3 If ρ is an operator

Write $\Delta = w' - w$. Then

$$i\big[e^{-ik\cdot\Delta}\mathcal{O} - e^{ik\cdot\Delta}\mathcal{O}^\dagger\big] = \sin(k\cdot\Delta)\,(\mathcal{O}+\mathcal{O}^\dagger) + \cos(k\cdot\Delta)\,i(\mathcal{O}-\mathcal{O}^\dagger). \tag{2.4}$$

Both $\mathcal{O}+\mathcal{O}^\dagger$ and $i(\mathcal{O}-\mathcal{O}^\dagger)$ are hermitian, so their expectation values are real. $\mathcal{O}-\mathcal{O}^\dagger$ consists of commutators, e.g. $[\rho^a(x),\rho^b(y)] = if^{abc}\rho^c(x)\delta^{(2)}(x-y)$. The resulting $i\cdot i$ gives real two-ρ terms. For classical ρ these commutators vanish and (2.3) holds.

3 Group I (add c.c.)

Row	class	remarks
II, N_1	R	real coefficient, 2ρ; with c.c. the kernel is symmetrised under $(x,w)\leftrightarrow(x',w')$
II, N_{2b}	I (implicit)	see below; rename the dummy $\int d^2k$ to $\int d^2q$ (it clashes with the external k)
III	I	explicit i; logs with sign-changing arguments: Sec. 6
IV	I	L_x has a positive argument
V, first part	I	phase $e^{-ik\cdot(w'-Y_p)}$ (see below)
V, second part	R	real coefficient, 2ρ

N_{2b}, implicit i. Define $\Phi^i \equiv \int d^2q\,d^2p\,e^{iq\cdot(w-y)}e^{ip\cdot(y-x)}\frac{R^i(q,p)}{q^2p^2}$. Every term of R^i is odd under $(q,p)\to(-q,-p)$: k^i or p^i times even functions. Hence $\Phi^{i*} = \int e^{-i(\dots)}\frac{R^i(q,p)}{q^2p^2} = \int e^{+i(\dots)}\frac{R^i(-q,-p)}{q^2p^2} = -\Phi^i$. So $\Phi^i = i\times(\text{real})$, and N_{2b} is class I.

Row V phases. $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)p^+/k^+} = e^{-ik\cdot(w'-Y_p)}$ with the real point $Y_p = (1-\frac{p^+}{k^+})z + \frac{p^+}{k^+}x$. So the c.c. gives $2\sin(k\cdot(w'-Y_p))$ in the first part and $2\cos(k\cdot(w'-Y_p))$ in the second.

Example (Row IV + c.c., classical ρ). $\frac{ig^4}{8\pi^5}e^{-ik\cdot(w'-w)}+\text{c.c.} \to \frac{g^4}{4\pi^5}\sin(k\cdot(w'-w))$, with everything else unchanged.

4 Group II (no c.c.): each row is hermitian by itself

4.1 Row I, first part: $N_{1b} = N_{1a}^\dagger$

Take $N_{1a}^\dagger$:

- $-i\to+i$ and $e^{-ik\cdot(w'-w)}\to e^{+ik\cdot(w'-w)}$;
- the two colour brackets change order; each is hermitian, since $\{\rho,\rho\}^\dagger = \{\rho,\rho\}$ and U, f are real;
- then relabel $w\leftrightarrow w'$ to restore the phase.

The kernel becomes $\frac{(x'-w)^i(y-z)^j}{\dots}$, the logs and bracket get $w\to w'$, and the colour factor becomes $[f\,U(w')\rho(x) - \dots]\,[U(w)\{\rho(x'),\rho(y)\} - \dots]$. This is *exactly* N_{1b}, including the $+i\log(1-\kappa)$ line. Hence $N_{1a}+N_{1b} = N_{1a}+N_{1a}^\dagger$ is real; for classical ρ it equals $-\frac{g^4}{8\pi^5}\frac{1}{k^+}\int\sin(k\cdot(w'-w))\,(\dots)$.

4.2 Row I, second part (N_{2a}–N_{2f})

All coefficients are real and all logs have positive arguments (L_x, $L_{x'}$, and the ratio in N_{2a}). Under $\sigma:(x,w)\leftrightarrow(x',w')$, with $D\equiv(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2\to -D$:

- N_{2a}: the bracket's terms pair up $1\leftrightarrow5$, $2\leftrightarrow7$, $3\leftrightarrow6$, $4\leftrightarrow8$. The log difference is odd, and so is $1/D$, so the product is even.
- N_{2b}: the first piece reduces to $\log(1-\kappa)\,[\dots]/((x-w)^2(x'-w')^2)$, which is even. The second and third pieces map into each other.
- N_{2c}, N_{2d}, N_{2e}, N_{2f}: in each, the two pieces map into each other ($L_x\leftrightarrow L_{x'}$), and each bracket is even. The middle piece of N_{2f} is even.

Checked numerically: every kernel equals its σ-image to machine precision.

The colour factor under σ followed by † (reversing $\rho(x')\rho(x)$): its first and fourth terms map to themselves, and the second and third map into each other after renaming dummy indices. So $N_{2\cdot}^{\dagger} = \int e^{+ik\cdot(w'-w)}\mathcal{K}\,\mathcal{O}^{\dagger} \overset{\sigma}{=} \int e^{-ik\cdot(w'-w)}\mathcal{K}\,\mathcal{O} = N_{2\cdot}$, i.e. hermitian. For classical ρ only $\cos(k\cdot(w'-w))$ survives.

The denominators $D=0$ and $(x-z)^2=(x-w)^2$ are removable singularities: there the attached logs vanish. For example, at $D=0$ the N_{2a} log difference is $\log\frac{[(1-\kappa)r+\kappa](x-w)^2}{[(1-\kappa)r+\kappa](x'-w')^2} - \log\frac{(x-w)^2}{(x'-w')^2} = 0$.

4.3 Row IV

Real coefficient. The kernel and the colour structure are symmetric under $(x,y)\leftrightarrow(x',y')$ with †, so it is real: for classical ρ, $e^{-ik\cdot(y'-y)/\xi}\to\cos(k\cdot(y'-y)/\xi)$.

5 Group III (add c.c.)

- Rows I, II, III (first part), V, VI are class I. Row III (second part) is class R.
- All denominators are positive, $p^+(x-z)^2+k^+(x-w)^2>0$, and all logs are L_x or $\log(1-\kappa)$.
- The phases combine into one real phase, $e^{-ik\cdot(y'-w)}e^{-ik\cdot(y'-z)p^+/k^+} = e^{-ik\cdot[(1+\frac{p^+}{k^+})y'-w-\frac{p^+}{k^+}z]}$.

So with c.c. each row is real: sin for class I, cos for class R.

6 Group I, Row III: the only genuine issue

Let $u\equiv\frac{(x-z)^2}{(x-w)^2}$ and $\kappa=\frac{k^+}{\vee}$. The logs in the file are

$$\begin{aligned}
&\ell_1 = \log\frac{u-\kappa}{u-1}\ \ (N_1) \qquad = \log\frac{\kappa-u}{1-u}\ \ (N_2), && \text{argument}<0 \text{ for } \kappa<u<1,\\
&\ell_2 = \log(1-u)\ \ (N_1,\ N_{3a}), && \text{argument}<0 \text{ for } u>1,\\
&\ell_3 = \log(\kappa-u)\ \ (N_1,\ \text{last line}), && \text{argument}<0 \text{ for } u>\kappa.
\end{aligned}$$

Origin. All three come from one p^+ integrand with the denominator $p^+(x-z)^2-k^+(x-w)^2$. Its zero $p^+_*=k^+/u$ lies in $(k^+,\vee)$ for $\kappa<u<1$ (giving ℓ_1) and in $(0,k^+)$ for $u>1$ (giving ℓ_2).

They always appear in fixed combinations.

- In N_1: the kernel $\mathcal{K}_1=\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\frac{(x-z)^2+(x-w)^2}{[(x-z)^2-(x-w)^2]^2}$ multiplies $-\log(1-\kappa)+\ell_1+\ell_2$.
- In N_2+N_{3a}: every kernel of N_2 that multiplies ℓ_1 has a partner in N_{3a} with the same kernel multiplying ℓ_2. Explicitly: (3) $\leftrightarrow$3rd, (4) $\leftrightarrow$5th, (6) $\leftrightarrow$4th, (7) $\leftrightarrow$6th. The double-pole kernel $\mathcal{N}=\frac{(z-w)\cdot[(x-z)(x-w)^2-(x-w)(x-z)^2]}{[(x-w)^2-(x-z)^2]^2}(\dots)$ multiplies $-\log(1-\kappa)+\ell_1+\ell_2$. So N_2+N_{3a} depends on the logs only through $\ell_1+\ell_2$.

Reading 1: principal value (logs $\to\log|\cdot|$). Then

$$\ell_1+\ell_2\to\log|u-\kappa|,\qquad \ell_3\to\log|u-\kappa|,\qquad -\log(1-\kappa)+\ell_1+\ell_2\to\log\frac{|u-\kappa|}{1-\kappa}\xrightarrow{u\to1}0. \tag{6.1}$$

The coefficient of every double pole $[(x-z)^2-(x-w)^2]^{-2}$ vanishes linearly on the line $|x-z|=|x-w|$, leaving a simple pole. A simple pole changes sign across the line, so the x integral exists (as a principal value). Row III is then *finite and real*: class I.

Reading 2: complex logs (principal branch, or any $\pm i\epsilon$). Then $\mathrm{Im}(\ell_1+\ell_2)=\pm\pi\,\theta(u-\kappa)$ and $\mathrm{Im}\,\ell_3=\pm\pi\,\theta(u-\kappa)$; there is no sign choice that makes them vanish near $u=1$, since ℓ_1 and ℓ_2 live on disjoint intervals. In T this produces $i\cdot i\pi=-\pi$, i.e. *real* terms. After adding the c.c. they become

$$\mp\frac{2\pi g^4}{16\pi^5}\frac{1}{k^+}\int\cos(k\cdot(w'-w))\,\theta(u-\kappa)\big[\mathcal{K}_1+\dots\big]\,f^{abd}\,\big[\dots\big]\big[\dots\big]. \tag{6.2}$$

These are real but *divergent*. At $(x-z)^2=(x-w)^2$, $\mathcal{K}_1\propto[(x-z)^2-(x-w)^2]^{-2}\sim t^{-2}$ (t is the distance from the line), with a nonzero numerator $[(x-z)^2+(x-w)^2]\,(x'-w')\cdot(y-z)$. Likewise $\mathcal{N}$ has the nonzero numerator $-(z-w)^2(x-w)^2$ there. So $\int dt\,t^{-2}$ diverges. The c.c. does not cancel this; it doubles it.

Conclusion for Row III.

- Reality alone does not decide the reading: the c.c. makes either reading real.
- *Finiteness* decides it. The p^+ pole must be taken as a principal value, i.e. every log in Row III is $\log|\cdot|$.
- With this reading the whole row depends only on $\log|u-\kappa| = \log\left|\frac{(x-z)^2-\kappa(x-w)^2}{(x-w)^2}\right|$ and $\log(1-\kappa)$.
- For $\vee \to \infty$ ($\kappa \to 0$) this is $\log\frac{(x-z)^2}{(x-w)^2}$, and $\log(1-\kappa) \to 0$: no negative argument is left anywhere.

No other row contains the $[(x-z)^2-(x-w)^2]^{-2}$ structure with an imaginary coefficient, so the $i\pi$ terms of Reading 2 could not be cancelled elsewhere in this file.

7 Summary

Source of i	where	how it disappears
explicit i (C1)	GI R3–R5, GII R1(1a,1b), GIII R1–R3, R5, R6	with c.c.: $i\,e^{-ik\cdot\Delta}$ + c.c. $= 2\sin(k\cdot\Delta)$; GII: $N_{1b} = N_{1a}^\dagger$
phases (C2)	all rows	with c.c. (GI, GIII) or σ-symmetry (GII): cos, sin
N_{2b} momentum integral (C3)	GI R2	$\Phi^i = i\times$ real, so class I
negative-argument logs (C4)	GI R3 only	principal value; all logs combine into $\log\|u-\kappa\|$
ρ ordering (C5)	3-ρ terms	$i(\mathcal{O}-\mathcal{O}^\dagger)$ is hermitian (commutators)

Changes needed in `Other_terms.tex`.

1. GI, Row III: write every log as $\log|\cdot|$, or equivalently replace $\ell_1+\ell_2 \to \log|u-\kappa|$ and $\ell_3 \to \log|u-\kappa|$.
2. GI, Row II, N_{2b}: rename the dummy $\int d^2k$ to $\int d^2q$.
3. State explicitly that Groups I and III are $T+T^\dagger$. The final real forms follow from (2.2) and (2.3).

Not checked here: the individual coefficients and kernels of each row. This note checks only the reality structure.